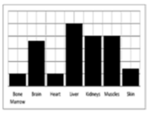
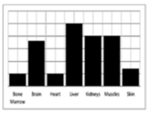
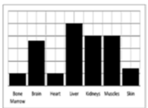
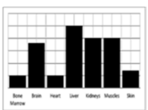
1.1.b. Cardiovascular and respiratory systems

Learners will know key terms and develop their knowledge and understanding of the cardiovascular and respiratory systems at rest, during exercise and during recovery.

Knowledge and understanding of the recovery system and how the body returns to its pre-exercise state will also be developed.

Learners understanding of the cardiovascular, respiratory and neuromuscular systems will also be applied to altitude training and exercise in the heat to show how these types of training can affect the body systems.

Topic Area	**Content**
Cardiovascular system at rest 	• the relationship between, and resting values for: ○ heart rate ○ stroke volume ○ cardiac output ○ methods of calculating the above • cardiac cycle: ○ diastole ○ systole • conduction system of the heart linked to the cardiac cycle.

Topic Area	**Content**
Cardiovascular system during exercise of differing intensities and during recovery  	• effects of different exercise intensisties and recovery on: ○ heart rate ○ stroke volume ○ cardiac output ○ methods of calculating the above • redistribution of cardiac output during exercise of differing intensities and during recovery: ○ vascular shunt mechanism ○ role of the vasomotor centre ○ role of arterioles ○ role of pre-capillary sphincters • mechanisms of venous return during exercise of differing intensities and during recovery • regulation of heart rate: ○ neural factors ○ hormonal factors ○ intrinsic factors.
Respiratory system at rest 	• relationship between resting values for: ○ breathing frequency ○ tidal volume ○ minute ventilation ○ methods of calculating the above • mechanics of breathing at rest and the muscles involved: ○ diaphragm ○ external intercostals ○ at the alveoli ○ at the muscles.
Respiratory system during exercise of differing intensities and during recovery  	• effects of differing intensities of exercise and recovery on: ○ breathing frequency ○ tidal volume ○ minute ventilation • mechanics of breathing during exercise of differing intensities and during recovery, including additional muscles involved: ○ inspiration – sternocleidomastoid, pectoralis minor ○ expiration – internal intercostals, rectus abdominis. • regulation of breathing during exercise of different intensities and during recovery ○ neural control ○ chemical control • effect of differing intensities of exercise and recovery on gas exchange at the alveoli and at the muscles ○ changes in pressure gradient ○ changes in dissociation of oxyhaemoglobin.